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Development of an Integrated System for Acoustic Calibration of Stethoscopes

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This report partially satisfies the requirements defined for the Project/Internship course, in the 3rd year, of the Degree in Electrical and Computer Engineering.

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Abstract

Aqui deverá ser apresentado o resumo de todo o trabalho efetuado. Esta secção não deverá exceder uma página.

Deve contextualizar o problema que pretende resolver ou a hipótese que irá formular, procure evidenciar as vantagens e desvantagens (se as houver) da solução encontrada, como também a forma através da qual a solução/hipótese foi validada. Neste último ponto, deverá referir-se aos desenvolvimentos efetuados, e à forma como validou (conformidade) e avaliou (desempenho) a solução encontrada.

O documento deve conter sempre duas versões do resumo: uma primeira no idioma do texto principal e a segunda num outro idioma. Este *template* assume que os dois idiomas em consideração são sempre Português e Inglês, assim, a classe irá colocar os cabeçalhos respetivos de acordo com o idioma selecionado nas opções da classe no ficheiro `main.tex`.

Keywords: Lista, separada por vírgulas, de palavras, frases, ou acrónimos chave no âmbito do trabalho descrito neste texto.

Resumo

The summary of all the developed work should be presented here. This section should not exceed one page.

Start the abstract with the contextualization of the problem you intend to solve or the hypothesis you will formulate. Try to highlight the advantages and disadvantages (if any) of the solution found, as well as the way in which the solution/hypothesis was validated. In this last point, you should refer to the developments made, and to the way you validated (compliance) and evaluated (performance) the solution found.

The document must always contain two versions of the abstract: a first in the language of the main text and the second one in another language. This template assumes that the two languages are always Portuguese and English, therefore, the class will place the correct section headers according to the language selected in the class options in the `main.tex` file.

Palavras-Chave: Comma separated list of words, phrases, or key acronyms within the scope of your developed work.

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Listings

List of Algorithms

List of Symbols

Symbol	Description	Units
f	força	N
i	corrente	A
M	massa	kg
P	potência	W
x	deslocamento	m
θ	deslocamento angular	rad
ω	velocidade angular	$\mathrm{rad\,s^{-1}}$

Chapter 1

Introduction

1.1 State of the Art/Contextualization

Stethoscopes remain the most fundamental and prevasive diagnostic tools in clinical practice for the auscultation of cardiopulmonary and vascular sounds. Currently, traditional acoustic stethoscopes dominate over 90% of the global market due to their high durability, low cost, and independence from electrical power sources [1]. However, purely acoustic auscultation is inherently limited not only by the fallibility of human auditory perception but also by the unrecognized physical degradation of the instrument over time. Studies have demonstrated that general practitioners miss up to 57% of significant valvular heart diseases when relying on standard stethoscopes [2].

While frequently attributed to human auditory limitations, this high error rate may be significantly composed by unnoticed material failures; such as compromised acoustic seals in the eartips, hardened tubing, or worn diaphragms. These physical degradations silently alter the stethoscope's frequency response, attenuating vital low-frequency biological sounds before they ever reach the clinician's ear. To mitigate these human diagnostic errors, the medical field is experiencing the gradual emergence of electronic stethoscopes designed to amplify, filter, and digitize biological sounds.

The clinical consequences of an uncalibrated or degraded stethoscope are severe. When physical degradation, such as hardened acoustic tubing or compromised eartip seals, artificially attenuates low-frequency biological signals, critical acoustic

biomarkers like the S3 gallop (a primary indicator of heart failure for older adults) may fall below the threshold of hearing. This mech nical attenuation directly leads to false-negative diagnoses, delayed medical interventions, and compromised patient safety. Therefore, establishing a reliable calibration system is not merely a matter of instrument maintenance, but a critical requirement for ensuring diagnostic integrity.

1.2 Project Description

This project proposes the development of an acoustic calibration system designed to characterize the frequency response, and signal fidelity of stethoscopes. The proposed architecture consists of two primary domains: a physical hardware module for sound generation and detection, and a software platform for control, data acquisition, and spectral analysis.

1.2.1 Objectives

By the end of the project, the developed system should be able to:

- **Simulate Cardiopulmonary Acoustics:** Generate controlled acoustic signals that mimic the frequency spectrum of biological sounds, particularly in the frequency range for cardiopulmonary auscultation.
- **Capture High-Fidelity Audio Data:** Accurately measure the stethoscope's output signal using a high-fidelity microphone and data acquisition system, ensuring that the measurement setup itself does not introduce significant distortions.
- **Analyze Frequency Response:** Analyze the acquired audio data to compute the stethoscope's frequency response curve utilizing python-based spectral analysis techniques.
- **Establish a Repeatable Protocol:** Develop a standardized testing protocol that can be easily replicated for routine stethoscope calibration in clinical settings, ensuring consistent and reliable performance assessments.

Chapter 2

Literature Review

2.1 Previous Work

Chapter 3

Methodology

3.1 System Requirements

To design an integrated calibration rig capable of characterizing acoustic and electronic stethoscopes, the system must satisfy a rigorous set of functional requirements (defining operational capabilities) and non-functional requirements (defining performance constraints and signal fidelity).

3.1.1 Functional Requirements

The functional requirements outline the necessary signal generation, mechanical coupling, and digital processing capabilities of the calibration rig, as detailed in Table 3.1.

Table 3.1: Functional requirements for the integrated system.

Requirement	**Engineering Description**
Controlled Acoustic Excitation	The system must generate continuous logarithmic sine sweeps covering the full cardiopulmonary frequency spectrum (20 Hz to 2000 Hz).
Structure-Borne Vibration	Acoustic energy must be delivered via a surface exciter mechanically coupled to an elastomeric tissue-simulating medium, avoiding free-air acoustic impedance mismatch.
Acoustic Sealing & Acquisition	The acquisition sensor must form an airtight seal with the stethoscope eartip to prevent low-frequency attenuation and accurately capture cardiopulmonary acoustics down to 20 Hz.
Signal Pre-Amplification	The low-amplitude analog output from the microphone capsule must be buffered and amplified to line-level voltages without clipping or harmonic distortion.
Digital Signal Processing (DSP)	The Central Control PC must acquire the digitized audio stream, compute the Fast Fourier Transform (FFT), and derive the system's frequency response curve.

3.1.2 Non-Functional Requirements

Non-functional requirements dictate the electronic precision, noise limits, and sampling criteria necessary to ensure clinical relevance and measurement repeatability, as presented in Table 3.2.

Table 3.2: Non-functional performance constraints and hardware specifications.

Requirement	**Engineering Description**
Low Total Harmonic Distortion	The power amplification stage must maintain high linearity across the target spectrum without introducing high-frequency switching noise, justifying a Class-AB analog topology.
Ultra-Low Input Noise Floor	The pre-amplifier stage must introduce minimal equivalent input noise ($e_n \leq 5$ nV/$\sqrt{\text{Hz}}$) to resolve faint biological sounds such as pulmonary crackles.
Nyquist-Compliant Sampling	The ADC stage must sample at a minimum rate of 44.1 kHz with at least 16-bit resolution to prevent quantization error and aliasing.
Measurement Repeatability	The physical coupler and elastomer medium must allow different stethoscope models to be tested under identical mechanical loading conditions.

3.2 System Architecture

To satisfy the functional and non-functional requirements established in Section 3.1, the calibration rig is divided into three interconnected domains: the *Sound Generation Stage*, the *Mechanical Domain*, and the *Sound Acquisition Stage*. Figure 3.1 illustrates the complete electromechanical signal path of the proposed architecture.

The signal path originates at the **Central Control PC**, where a digital reference waveform is converted to an analog signal (V_{out}) via a digital-to-analog converter (DAC). This signal drives a custom discrete Class-AB power amplifier based on the TDA2030A integrated circuit, which provides the linear current necessary to excite the Dayton Audio DAEX25 surface exciter without introducing high-frequency switching artifacts.

Within the *Mechanical Domain*, the exciter transfers structure-borne vibrations through a silicone elastomer phantom into the stethoscope's chestpiece. The acoustic wave travels through the stethoscope's internal acoustic tube to the eartip, where it is captured by a -35 dB Electret Condenser Microphone (ECM) capsule housed within a compliant physical coupler designed to ensure an airtight acoustic seal.

Finally, in the *Sound Acquisition Stage*, the microvolt-level signal produced by the ECM capsule is conditioned by an ultra-low-noise NE5532P pre-amplifier circuit. The boosted line-level signal is digitized by a dedicated 16-bit USB Analog-to-Digital Converter and transmitted back to the Central Control PC for Fast Fourier Transform (FFT) and spectral analysis.

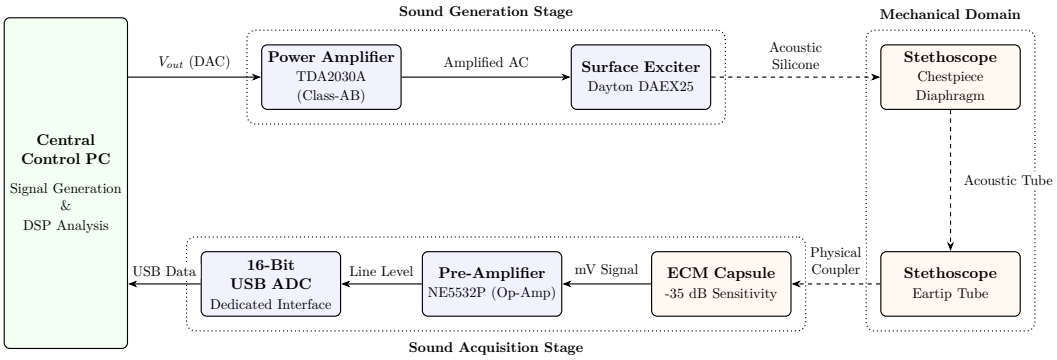


Figure 3.1: Block diagram of the proposed physical rig

3.3 Component Selection

3.3.1 Sound Generation Technologies

Traditional audio testing frequently relies on standard dynamic speakers. However, dynamic speakers operate by displacing free air.

Because a medical stethoscope is clinically designed to capture vibrations through high-impedance solid mediums (such as human tissue and fluid), utilizing an air-coupled speaker introduces significant energy loss, particularly in the low-frequency cardiovascular bands (20 Hz - 150 Hz).

To overcome this limitation, modern biomedical testing methodologies require the use of surface transducers or exciters. Instead of pushing air, surface transducers are designed to directly vibrate the solid object to which they are attached. When a surface transducer is mechanically coupled with a tissue-simulating medium (such as silicone or acoustic gelatine), it can accurately replicate the acoustic impedance of a human chest [3].

Following the justification for structure-borne sound converters, three commercially available surface transducers were selected for evaluation in this project:

- **Dayton Audio TT25-8 PUCK:** This is a compact, high-performance tactile transducer designed for low-frequency applications. It can produce strong vibrations in the 20 Hz to 80 Hz range and comes at a price range of around 19€.
- **Dayton Audio HDN-8 Weatherproof:** This is a high-performance, weatherproof tactile transducer designed for outdoor use. It can produce strong vibrations in the 20 Hz to 80 Hz range and comes at a price range of around 25€.
- **Dayton Audio DAEX25FHE-4:** This is a compact and affordable exciter with a frequency response of 70 Hz to 20 kHz and comes at a price range of around 15€.

For the sake of this project, the **Dayton Audio DAEX25FHE-4 exciter** was selected as the most suitable option for our stethoscope calibration system. This choice was based on its ability to produce strong vibrations in the relevant frequency range for cardiopulmonary sounds, as well as its affordability and compact size, which are important considerations for a practical calibration system.

Power Amplifier

To drive the Dayton Audio DAEX25FHE-4 exciter, the reference signal generated by the central control software must be passed through a power amplifier. The primary requirement for this component is signal linearity; the amplifier must deliver a perfectly flat frequency response from 20 Hz to 2000 Hz without introducing Total Harmonic Distortion (THD) or low-frequency roll-off.

Two primary amplifier topologies were evaluated for the physical test rig:

- **Class-D Amplifiers (e.g., PAM8403 / TPA3116):** Class-D modules operate utilizing PWM. While extremely affordable and capable of operating on

standard 5V USB power, the digital switching nature of Class-D topologies introduces high-frequency switching noise and poorer inherent linearity [4].

- **Class-AB Amplifiers (e.g., TDA2030A / LM1875):** Class-AB amplifier provides a continuous, highly linear analog output, ensuring a pure analog signal path across the entire target frequency spectrum. Literature demonstrates that Class-AB topologies consistently outperform Class-D stages in Total Harmonic Distortion (THD) metrics, frequently achieving distortion figures up to 30 dB better than their digital counterparts [5].

To ensure the acoustic reference signal remained free of high-frequency digital artifacts and low-frequency attenuation, the **Class-AB TDA2030A** IC was selected. It comes at a highly cost-effective price point of around 2€.

The TDA2030A was then integrated into a power amplifier circuit to provide sufficient current to drive the Dayton Audio DAEX25FHE-4 exciter without introducing clipping or harmonic distortion.

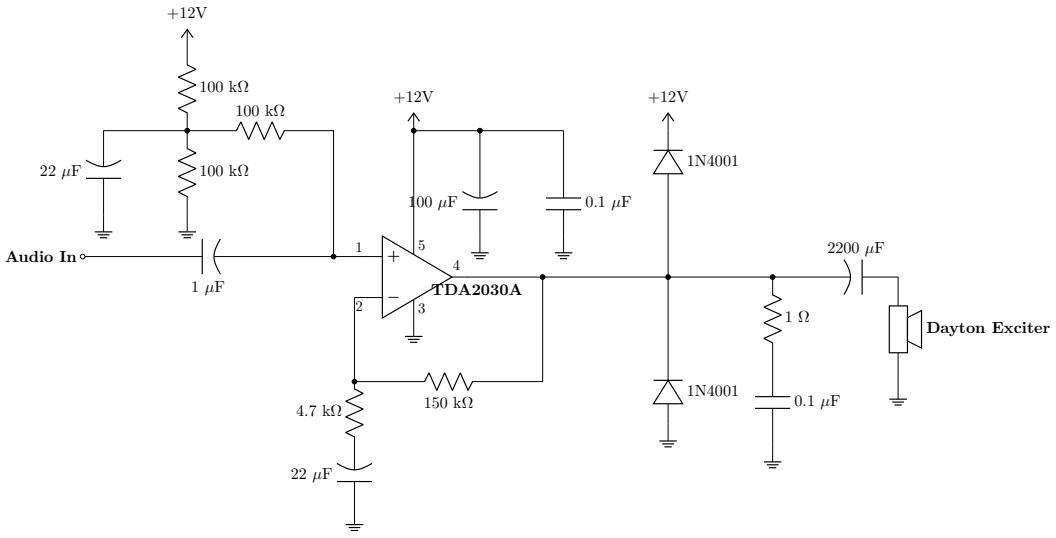


Figure 3.2: Final layout of the custom TDA2030A power amplifier, featuring perfect dynamic coordinate alignment and a spacious decoupling layout.

3.3.2 Sound Acquisition Technologies

To accurately capture the acoustic output of the stethoscope, the microphone must satisfy two critical requirements: it must possess a physical form factor small enough to achieve an airtight seal with the stethoscope's eartip (approximately 5 mm in diameter), and it must feature a flat frequency response capable of registering extreme low-frequency cardiovascular sounds down to 20 Hz.

Two primary microphone technologies were evaluated for the acquisition stage:

- **MEMS (Micro-Electro-Mechanical Systems) Microphones:** Commonly utilized in modern mobile devices, MEMS microphones are exceptionally compact and digitally integrated. However, consumer-grade MEMS capsules are specifically engineered for human speech transmission. Consequently, they frequently incorporate built-in high-pass filters designed to attenuate wind and handling noise below 60 Hz. Utilizing a standard MEMS microphone would result in the severe attenuation of foundational S1, S2, and S3 heart sounds, rendering them unsuitable for this calibration system.
- **ECM (Electret Condenser Microphone) Capsules:** ECMs are traditional analog acoustic sensors utilized in audio measurement. A standard 6 mm ECM capsule provides the necessary compact form factor for an airtight eartip coupling. More importantly, unlike consumer MEMS devices, measurement-grade ECM capsules do not artificially filter low-frequency kinetic energy, providing a highly linear frequency response down to 20 Hz.

Given the low-frequency requirements of cardiopulmonary auscultation, a 6 mm ECM capsule was selected as the optimal sensor for the acquisition stage. Because the raw analog output of an ECM capsule is highly attenuated (typically in the millivolt range), this selection necessitates the integration of a low-noise pre-amplification stage prior to digital signal processing.

Table 3.3: Conversion of Microphone Sensitivity (dBV/Pa) to Raw Voltage Output (mV/Pa)

Microphone Example	**Sensitivity (dBV/Pa)**	**Voltage Output (mV/Pa)**
Generic Low-Cost ECM	-60 dB	1.00 mV
High-Sensitivity ECM	-40 dB	10.00 mV
Selected Prototype ECM	**-35 dB**	**17.78 mV**
Premium Measurement Mic	-20 dB	100.00 mV

As demonstrated in Table 3.3, microphone sensitivity operates on a logarithmic scale. A Generic Electret Condenser Microphone rated at -60 dB outputs approximately 1.00 millivolts per Pascal of acoustic pressure. By acquiring a higher sensitivity capsule rated at -35 dB, the electrical output increases drastically to 17.78 millivolts per Pascal, yielding a signal almost 18 times stronger for the exact same acoustic event.

This stronger signal reduces the gain requirements for the subsequent ADC preamplification stage, therefore minimizing the amplification of inherent electronic noise and improving the overall Signal-to-Noise Ratio (SNR) of the acquired data.

Pre-Amplifier

Despite selecting a high-sensitivity Electret Condenser Microphone (-35 dBV/Pa), the raw analog output generated by cardiopulmonary acoustics remains in the low millivolt range. To prevent quantization error and signal loss during analog-to-digital conversion, the microphone's output must be conditioned and boosted using a dedicated pre-amplifier stage. The selection of the Operational Amplifier (Op-Amp) for this stage is highly critical, as any inherent electronic noise introduced here will be amplified and permanently embedded into the acquired acoustic data.

Three pervasive operational amplifiers were evaluated for this signal conditioning stage:

- **LM358 (General Purpose Op-Amp):** While exceptionally affordable and common in basic microcontroller kits, the LM358 is a general-purpose operational amplifier not intended for high-fidelity audio applications. It suffers from significant crossover distortion and possesses a relatively high input noise profile, making it unsuitable for capturing delicate, low-amplitude respiratory frequencies.

- **TL072 (JFET-Input Op-Amp):** The TL072 is a widely utilized, audio-grade Op-Amp commonly found in electronic stethoscope designs. Its JFET inputs provide high input impedance, and it boasts a low Total Harmonic Distortion (THD) profile alongside a low noise floor (typically 18 $\text{nV}/\sqrt{\text{Hz}}$). It is an excellent, low-cost option for general biological signal amplification.

- **NE5532P (Ultra-Low Noise Audio Op-Amp):** The NE5532 is historically considered the industry standard for professional audio pre-amplification. It features a high slew rate and an ultra-low noise floor (typically 5 $\text{nV}/\sqrt{\text{Hz}}$). While it draws slightly more quiescent current than the TL072, it offers superior linear performance and exceptionally clean gain across the entire 20 Hz to 20 kHz spectrum.

Table 3.4: Datasheet Specification Comparison: NE5532P vs. TL072

Parameter	**NE5532P**	**TL072**	**Relevance to Calibration Rig**
Input Architecture	BJT	JFET	JFETs offer higher input impedance, but BJTs offer lower voltage noise.
Equivalent Input Noise	5 nV/$\sqrt{\text{Hz}}$	18 nV/$\sqrt{\text{Hz}}$	NE5532 is significantly quieter, crucial for capturing faint respiratory crackles without static.
Gain Bandwidth (GBW)	10 MHz	3 MHz	Both exceed the 20 kHz requirement, but NE5532 offers more linear gain at higher frequencies.
Slew Rate	9 V/μs	13 V/μs	Both are more than sufficient to accurately track rapid acoustic transients (e.g., S1/S2 heart valves snapping shut).

As detailed in Table 3.4, both operational amplifiers possess technical specifications that easily satisfy the baseline requirements for cardiopulmonary acoustic processing.

The primary challenge of this acquisition stage is the noise mitigation when amplifying the microvolt-level signals of faint respiratory anomalies. The NE5532P reduces the equivalent input noise down to 5 nV/$\sqrt{\text{Hz}}$, less than a third of the noise generated by the TL072. Given the necessity to capture low-amplitude anomalies, such as pulmonary crackles, without introducing electronic interference, the **NE5532P** was selected as the optimal Op-Amp for the pre-amplifier circuit. It comes at a highly cost-effective price point of around 1€.

The NE5532P was then integrated into a pre-amplihadwarefier circuit to boost the millivolt-level output of the ECM capsule to line-level voltages suitable for digitization by the ADC.

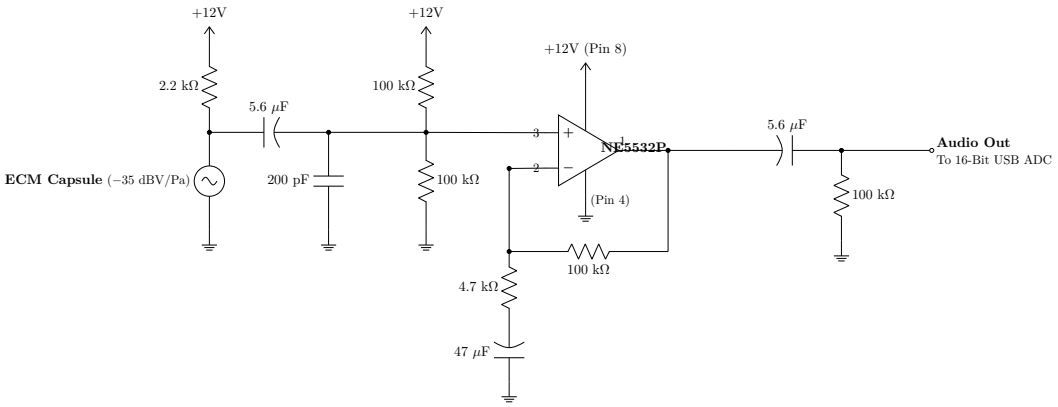


Figure 3.3: Schematic of the custom single-supply NE5532P audio preamplifier stage. The circuit features a classic non-inverting topology with 200 pF RF input filtering, single-supply mid-rail biasing, a 27 dB AC gain loop (100 kΩ/4.7 kΩ), and DC-blocked line-level audio conditioning.

3.3.3 Signal Digitization: Analog-to-Digital Conversion (ADC)

Following the analog pre-amplification stage, the signal must be digitized for computational analysis. The fidelity of the resulting Fast Fourier Transform (FFT) algorithms is heavily dependent on the resolution and sampling rate of the chosen Analog-to-Digital Converter (ADC).

Two primary digital conversion architectures were evaluated:

- **Microcontroller ADC (e.g., Arduino Uno 10-bit):**
- **Dedicated Audio ADC (16-bit USB Interface):**

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